

> 2 point of view for $\tau \frac{dV}{dt} = -V + 1$

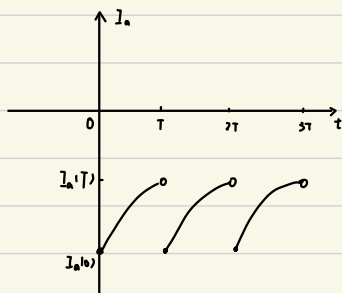
- from analytic solution: $V(t) = 1 - e^{-\frac{t}{\tau}}$
- from numerical solution: $\tau(V^{(k+1)} - V^{(k)}) = (1 - V^{(k)})\Delta t$
 - if $\tau < 1$ then when $| \geq V^{(k)} \geq 1$, $V^{(k+1)} < V^{(k)}$
 - if $\tau > 1$ then when $| \geq V^{(k)}$, $V^{(k+1)} > V^{(k)}$

> without I_a

- $T = -\tau \ln(1 - \frac{1}{2})$
- $f = \frac{1}{\tau \ln(1 + \frac{1}{2})} \sim \frac{1}{\tau}$ $f_{\tau} = \frac{1}{\tau \ln(1 + \frac{1}{2}) + \tau} \sim \frac{1}{\tau}$

> with I_a

- I_a : $I_a = -J_a \sum_{n=0}^{+\infty} e^{-\frac{nT+t}{\tau_a}} \quad t \in [0, T) = -J_a \left(\sum_{n=0}^{+\infty} e^{-\frac{nT}{\tau_a}} \right) e^{-\frac{t}{\tau_a}} = (-J_a) \left(\frac{1}{1 - e^{-T/\tau_a}} \right) e^{-\frac{t}{\tau_a}} \quad (\star_1)$



// $I_a(T) - I_a(0) = J_a$, which is independent of I_a

- simulate: $\tau_a \frac{dI_a}{dt} = -I_a - J_a \tau_a \sum \delta(t - t_i^{\text{spikes}})$

$\Downarrow$

$$\tau_a (I_a^{(k+1)} - I_a^{(k)}) = (-I_a^{(k)}) \Delta t - J_a \tau_a \cdot (0 \text{ or } 1)$$

$\Downarrow$

$$I_a^{(k+1)} = I_a^{(k)} \left(1 - \frac{\Delta t}{\tau_a} \right) - J_a \cdot (0 \text{ or } 1) \quad (\star_3)$$

- analytic // $\tau \frac{dV}{dt} = -V + I_e \Rightarrow V(t) = C e^{-\frac{t}{\tau}} + I_e$

$$\tau \frac{dV}{dt} = -V + (I_e + I_a) \Rightarrow V(t) = C e^{-\frac{t}{\tau}} + I_e + \frac{1}{\tau} e^{-\frac{t}{\tau}} \int I_a e^{+\frac{t}{\tau}} dt \quad t \in [0, T] = C e^{-\frac{t}{\tau}} + I_e + (-J_a) \left(\frac{1}{1 - e^{-T/\tau_a}} \right) \left(\frac{\tau_a}{\tau_a - \tau} \right) e^{-\frac{t}{\tau}}$$

$$\begin{cases} V(0) = 0 \\ V(T) = 1 \end{cases} \Rightarrow \frac{-1 - J_a (e^{\frac{T}{\tau_a} - 1})}{e^{\frac{T}{\tau}} - e^{-\frac{T}{\tau}}} + I_e + (-J_a) \left(\frac{1}{1 - e^{-T/\tau_a}} \right) \left(\frac{\tau_a}{\tau_a - \tau} \right) = 0$$

$$\Rightarrow -1 + I_e \left(1 - e^{-\frac{T}{\tau}} \right) + (-J_a) \left(\frac{1}{1 - e^{-T/\tau_a}} \right) \left(\frac{\tau_a}{\tau_a - \tau} \right) (e^{-\frac{T}{\tau_a}} - e^{-\frac{T}{\tau}}) = 0 \quad (\star_2) \quad // \text{pass dimension check!}$$

- In summary

- numerical solution of ODE: $\star_3$

- analytic solution of ODE: $V(t) = \left(\frac{-1 - J_a (e^{\frac{T}{\tau_a} - 1})}{e^{\frac{T}{\tau}} - e^{-\frac{T}{\tau}}} \right) e^{-\frac{t}{\tau}} + I_e + (-J_a) \left(\frac{1}{1 - e^{-T/\tau_a}} \right) \left(\frac{\tau_a}{\tau_a - \tau} \right) e^{-\frac{t}{\tau_a}} \quad t \in [0, T)$

- numerical solution of the algebraic equation: $I_e = I_e(T)$ ✓ $T = T(I_e)$ ✗

- analytic solution of the algebraic equation: $I_e = I_e(T)$ ✓ $T = T(I_e)$ ✓

- causality

$$I_e, \tau, \tau_a, J_a \Rightarrow T \quad (\star_2)$$

$$I_e, J_a, T \Rightarrow I_a(0), I_a(T) \quad (\star_1)$$